

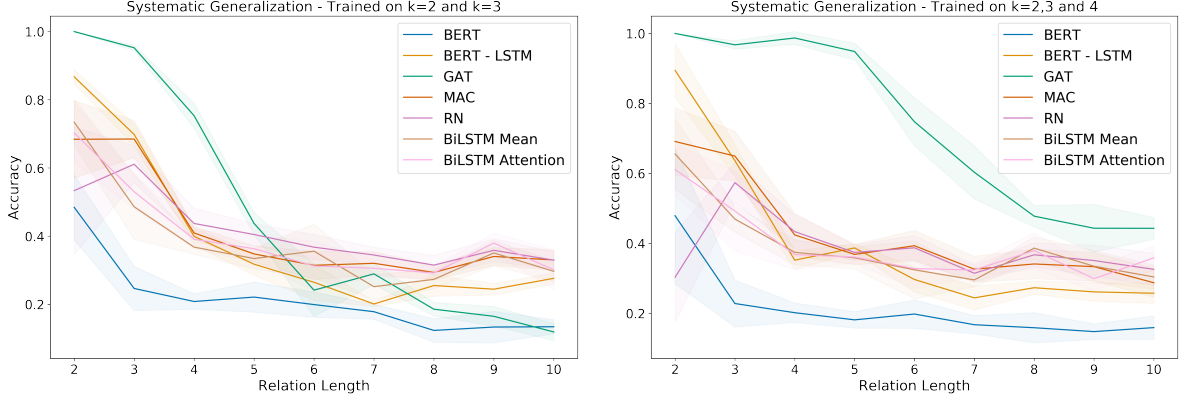


Figure 5: Systematic generalization performance of different models when trained on clauses of length $k = 2, 3$ (Left) and $k = 2, 3, 4$ (Right).

To make a prediction about a target query given a story, we concatenate the embedding of the story (generated by the baseline model) with the embeddings of the two target entities and we feed this concatenated embedding to a 2-layer feed-forward neural network with a softmax prediction layer.

4.2 Experimental Setup

Hyperparameters. We selected hyperparameters for all models using an initial grid search on the systematic generalization task (described below). All models were trained for 100 epochs with Adam optimizer and a learning rate of 0.001. The Appendix provides details on the selected hyperparameters.

Generated datasets. For all experiments, we generated datasets with 10-15k training examples. In many experiments, we report training and testing results on stories with different clause lengths k . (For brevity, we use the phrase “clause length” throughout this section to refer to the value $k = |\mathcal{B}_C|$, i.e., the number of steps of reasoning that are required to predict the target query.) In all cases, the training set contains 5000 train stories per k value, and, during testing, all experiments use 100 test stories per k value. All experiments were run 10 times with different randomly generated stories, and means and standard errors over these 10 runs are reported. As discussed in Section 3.5, during training we holdout 20% of the paraphrases, as well as 10% of the possible logical clauses.

4.3 Results and Discussion

With our experimental setup in place, we now address the three key questions (**Q1-Q3**) outlined at the beginning of Section 4.

bedding approaches.

Q1: Systematic Generalization

We begin by using CLUTRR to evaluate the ability of the baseline models to perform systematic generalization (**Q1**). In this setting, we consider two training regimes: in the first regime, we train all models with clauses of length $k = 2, 3$, and in the second regime, we train with clauses of length $k = 2, 3, 4$. We then test the generalization of these models on test clauses of length $k = 2, \dots, 10$.

Figure 5 illustrates the performance of different models on this generalization task. We observe that the GAT model is able to perform near-perfectly on the held-out logical clauses of length $k = 3$, with the BERT-LSTM being the top-performer among the text-based models but still significantly below the GAT. Not surprisingly, the performance of all models degrades monotonically as we increase the length of the test clauses, which highlights the challenge of “zero-shot” systematic generalization (Lake and Baroni, 2018; Sodhani et al., 2018). However, as expected, all models improve on their generalization performance when trained on $k = 2, 3, 4$ rather than just $k = 2, 3$ (Figure 5, right). The GAT, in particular, achieves the biggest gain by this expanded training.

Q2: The Benefit of Structure

The empirical results on systematic generalization also provide insight into how the text-based NLU systems compare against the graph-based GAT model that has full access to the logical graph structure underlying the stories (**Q2**). Indeed, the relatively strong performance of the GAT model (Figure 5) suggests that the language-based models fail to learn a robust mapping from the natural language narratives to the underlying logical facts.

Table 2: Testing the robustness of the various models when training and testing on stories containing various types of noise facts. The types of noise facts (supporting, irrelevant, and disconnected) are defined in Section 3.5.

Models		Unstructured models (no graph)						Structured model (with graph)
Training	Testing	BiLSTM - Attention	BiLSTM - Mean	RN	MAC	BERT	BERT-LSTM	GAT
Clean	Clean	0.58 $\pm$ 0.05	0.53 $\pm$ 0.05	0.49 $\pm$ 0.06	0.63 $\pm$ 0.08	0.37 $\pm$ 0.06	0.67 $\pm$ 0.03	1.0 $\pm$ 0.0
	Supporting	0.76 $\pm$ 0.02	0.64 $\pm$ 0.22	0.58 $\pm$ 0.06	0.71 $\pm$ 0.07	0.28 $\pm$ 0.1	0.66 $\pm$ 0.06	0.24 $\pm$ 0.2
	Irrelevant	0.7 $\pm$ 0.15	0.76 $\pm$ 0.02	0.59 $\pm$ 0.06	0.69 $\pm$ 0.05	0.24 $\pm$ 0.08	0.55 $\pm$ 0.03	0.51 $\pm$ 0.15
	Disconnected	0.49 $\pm$ 0.05	0.45 $\pm$ 0.05	0.5 $\pm$ 0.06	0.59 $\pm$ 0.05	0.24 $\pm$ 0.08	0.5 $\pm$ 0.06	0.8 $\pm$ 0.17
Supporting	Supporting	0.67 $\pm$ 0.06	0.66 $\pm$ 0.07	0.68 $\pm$ 0.05	0.65 $\pm$ 0.04	0.32 $\pm$ 0.09	0.57 $\pm$ 0.04	0.98 $\pm$ 0.01
Irrelevant	Irrelevant	0.51 $\pm$ 0.06	0.52 $\pm$ 0.06	0.5 $\pm$ 0.04	0.56 $\pm$ 0.04	0.25 $\pm$ 0.06	0.53 $\pm$ 0.06	0.93 $\pm$ 0.01
Disconnected	Disconnected	0.57 $\pm$ 0.07	0.57 $\pm$ 0.06	0.45 $\pm$ 0.11	0.4 $\pm$ 0.1	0.17 $\pm$ 0.05	0.47 $\pm$ 0.06	0.96 $\pm$ 0.01
Average		0.61 $\pm$ 0.08	0.59 $\pm$ 0.08	0.54 $\pm$ 0.07	0.61 $\pm$ 0.06	0.30 $\pm$ 0.07	0.56 $\pm$ 0.05	0.77 $\pm$ 0.09

To further confirm this trend, we ran experiments with modified train and test splits for the text-based models, where the same set of natural language paraphrases were used to construct the narratives in both the train and test splits (see Appendix 1.3 for details). In this simplified setting, the text-based models must still learn to reason about held-out logical patterns, but the difficulty of parsing the natural language is essentially removed, as the same natural language paraphrases are used during testing and training. We found that the text-based models were competitive with the GAT model in this simplified setting (Appendix Figure 1), confirming that the poor performance of the text-based models on the main task is driven by the difficulty of parsing the unseen natural language narratives.

Q3: Robust Reasoning

Finally, we use CLUTRR to systematically evaluate how various baseline neural language understanding systems cope with noise (Q3). In all the experiments we provide a combination of $k = 2$ and $k = 3$ length clauses in training and testing, with noise facts being added to the train and/or test set depending on the setting (Table 2). We use the different types of noise facts defined in Section 3.5.

Overall, we find that the GAT baseline outperforms the unstructured text-based models across most testing scenarios (Table 2), which showcases the benefit of a structured feature space for robust reasoning. When training on clean data and testing on noisy data, we observe two interesting trends that highlight the benefits and shortcomings of the various model classes:

1. All the text-based models excluding BERT actually perform better when testing on examples that have *supporting* or *irrelevant* facts added. This suggests that these models actually benefit from having more content related to the entities in the story. Even though this content is

not strictly useful or needed for the reasoning task, it may provide some linguistic cues (e.g., about entity genders) that the models exploit. In contrast, the BERT-based models do not benefit from the inclusion of this extra content, which is perhaps due to the fact that they are already built upon a strong language model (e.g., that already adequately captures entity genders.)

2. The GAT model performs poorly when *supporting* facts are added but has no performance drop when *disconnected* facts are added. This suggests that the GAT model is sensitive to changes that introduce cycles in the underlying graph structure but is robust to the addition of noise that is disconnected from the target entities.

Moreover, when we trained on noisy examples, we found that only the GAT model was able to consistently improve its performance (Table 2). Again, this highlights the performance gap between the unstructured text-based models and the GAT.

5 Conclusion

In this paper we introduced the CLUTRR benchmark suite to test the systematic generalization and inductive reasoning capabilities of NLU systems. We demonstrated the diagnostic capabilities of CLUTRR and found that existing NLU systems exhibit relatively poor robustness and systematic generalization capabilities—especially when compared to a graph neural network that works directly with symbolic input. These results highlight the gap that remains between machine reasoning models that work with unstructured text and models that are given access to more structured input. We hope that by using this benchmark suite, progress can be made in building more compositional, modular, and robust NLU systems.

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